

Wireshark: The 80/20 Guide for Kerberos Analysis

Master the 20% That Gives You 80% of the Value

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Focus: Analyzing Kerberos attacks for security research

Time to Competence: 45 minutes

Core Concept

Wireshark analyzes packet captures. For your Kerberos project, you need to:

1. Open pcap files from tcpdump
2. Find attack patterns
3. Extract evidence
4. Create screenshots for portfolio

That's it. Everything else is bonus.

Quick Start: Open Your First Capture

Step 1: Launch Wireshark

```
wireshark bruteforce_attack.pcap &
```

What you see:

- Packet list (top) - every packet
 - Packet details (middle) - decoded structure
 - Packet bytes (bottom) - raw data
-

Step 2: Apply Kerberos Filter

In the filter bar, type:

```
kerberos
```

Press Enter.

Result: Only Kerberos packets shown

The 5 Essential Filters

These cover 95% of Kerberos attack analysis:

Filter 1: Show All Kerberos

```
kerberos
```

Filter 2: Show Failed Authentications

```
kerberos.error_code == 24
```

24 = KDC_ERR_PREAUTH_FAILED (wrong password)

Filter 3: Show Successful Authentications

```
kerberos.msg_type == 11 && !kerberos.error_code
```

11 = AS-REP (authentication response) without error

Filter 4: Show AS-REQ (initial auth)

```
kerberos.msg_type == 10
```

Filter 5: Show TGS-REQ (service tickets)

```
kerberos.msg_type == 12
```



Analyzing Your Attack Captures

Attack 1: Brute Force Analysis

Open your brute force capture:

```
wireshark bruteforce_capture.pcap
```

Step 1: Apply error filter:

```
kerberos.error_code == 24
```

What to look for:

- Multiple packets from same IP
- Same username (cname)
- Regular timing pattern

Step 2: Count failed attempts:

- Statistics > Protocol Hierarchy
- Look at "Kerberos" line
- Note packet count

Step 3: Find source IP:

```
kerberos.error_code == 24
```

- Right-click any packet
- Follow > TCP Stream
- Note source IP in top

Step 4: Check if attack succeeded:

```
kerberos.msg_type == 11 && !kerberos.error_code && ip.src == 192.168.88.10
```

Replace with actual attacker IP

Evidence to capture:

- Screenshot of repeated failures
 - Statistics showing count
 - Timeline of attack
 - Successful auth (if any)
-

Attack 2: AS-REP Roasting Analysis

Open your AS-REP roasting capture:

```
wireshark asrep_roasting.pcap
```

Step 1: Find AS-REP without pre-auth:

```
kerberos.msg_type == 11
```

Step 2: Look for multiple different usernames:

- Click on packets
- Expand: Kerberos > as-rep > cname
- Note different usernames being tested

Step 3: Find successful roasts:

- Look for AS-REP responses (msg_type 11)
- These contain encrypted hashes
- Without error codes = roastable account found

Evidence to capture:

- Multiple AS-REP responses
- Different usernames tested
- Hash in encrypted portion (expand enc-part)

Key indicator:

```
kerberos.msg_type == 11 && !kerberos.pa_data
```

AS-REP without pre-auth data = VULNERABLE

Attack 3: Kerberoasting Analysis

Open your Kerberoasting capture:

```
wireshark kerberoasting.pcap
```

Step 1: Show service ticket requests:

```
kerberos.msg_type == 12
```

Step 2: Count unique services:

- Statistics > Conversations
- TCP tab
- Note multiple connections

Step 3: Extract service names:

- Filter: `kerberos.msg_type == 12`
- Expand: Kerberos > tgs-req > sname
- Note each service being requested

Evidence to capture:

- Multiple TGS-REQ packets
- Different service names
- Same source IP
- Short time window

Attack signature:

```
kerberos.msg_type == 12
```

Look for 10+ requests in short time from one IP

Essential Analysis Techniques

Technique 1: Follow TCP Stream

Purpose: See entire conversation

How:

1. Right-click any packet
2. Follow > TCP Stream
3. See full exchange

Use case: Understand attack sequence

Technique 2: Statistics > Conversations

Purpose: Find high-traffic sources

How:

1. Statistics > Conversations
2. TCP tab
3. Sort by "Packets" column

Use case: Identify attacker IP quickly

Technique 3: Time Analysis

Purpose: Calculate attack rate

How:

1. View > Time Display Format > Seconds Since Previous Packet
2. Look at time between packets

Use case: Measure brute force speed

Technique 4: Expert Information

Purpose: Find problems automatically

How:

1. Analyze > Expert Information
2. Look for warnings/errors

Use case: Quick anomaly detection

Technique 5: Protocol Hierarchy

Purpose: Understand traffic composition

How:

1. Statistics > Protocol Hierarchy
2. See breakdown of protocols

Use case: Verify capture contains expected traffic



Wireshark Display Filters Cheat Sheet

Kerberos Message Types

```
kerberos.msg_type == 10    # AS-REQ (initial authentication request)
kerberos.msg_type == 11    # AS-REP (authentication response)
kerberos.msg_type == 12    # TGS-REQ (service ticket request)
kerberos.msg_type == 13    # TGS-REP (service ticket response)
kerberos.msg_type == 30    # KRB-ERROR (error message)
```

Error Codes

```
kerberos.error_code == 6   # Client unknown (user doesn't exist)
kerberos.error_code == 24  # Pre-auth failed (wrong password)
kerberos.error_code == 25  # Pre-auth required (checking if needed)
```

Specific Users

```
kerberos.cname contains "admin"    # Client name contains "admin"
kerberos.sname contains "http"     # Service name contains "http"
```

Combinations

```
# Failed brute force attempts
kerberos.error_code == 24 && kerberos.cname contains "admin"
```

```
# Successful authentication
kerberos.msg_type == 11 && !kerberos.error_code

# Service ticket requests for HTTP
kerberos.msg_type == 12 && kerberos.sname contains "http"

# From specific IP
kerberos && ip.src == 192.168.88.10

# To specific IP (KDC)
kerberos && ip.dst == 192.168.88.20
```

Creating Portfolio Screenshots

Screenshot 1: Attack Overview

Purpose: Show scale of attack

Setup:

1. Apply filter: `kerberos.error_code == 24`
2. Statistics > Protocol Hierarchy
3. Screenshot showing packet count

Annotation:

```
"247 failed authentication attempts from 192.168.88.10
Target: admin@LAB.LOCAL
Duration: 3 minutes 45 seconds"
```

Screenshot 2: Attack Timeline

Purpose: Show attack pattern

Setup:

1. Filter: `kerberos.error_code == 24`
2. View > Time Display Format > Time of Day
3. Sort by time
4. Screenshot showing pattern

Annotation:

```
"Regular timing pattern indicates automated attack  
Average: 1.1 attempts per second"
```

Screenshot 3: Successful Compromise

Purpose: Show attack succeeded

Setup:

1. Filter: `kerberos.msg_type == 11 && !kerberos.error_code && ip.src == ATTACKER_IP`
2. Expand first success packet
3. Show enc-part (encrypted ticket)

Annotation:

```
"Attack successful after 234 attempts  
Ticket issued: 2025-02-07 10:25:47  
Valid for: 10 hours"
```

Screenshot 4: Attack Comparison

Purpose: Show before/after

Setup:

1. Open two Wireshark windows
2. Left: Normal traffic
3. Right: Attack traffic

Annotation:

```
"Normal: 10-20 packets/minute  
Attack: 247 packets in 3 minutes"
```

Essential Menu Actions

File Menu

```
File > Open                # Open pcap file
File > Export Packet Dissections > As Plain Text
                              # Export to text for reports
File > Export Specified Packets
                              # Save filtered packets to new pcap
```

Edit Menu

```
Edit > Find Packet        # Search for specific data
Edit > Find Next           # Continue search
```

View Menu

```
View > Time Display Format > Time of Day
                              # Human-readable timestamps
View > Coloring Rules        # Highlight packet types
View > Expand All            # See all packet details
```

Statistics Menu

```
Statistics > Protocol Hierarchy
                              # Traffic breakdown
Statistics > Conversations
                              # Connection pairs
Statistics > Endpoints
                              # Individual hosts
Statistics > I/O Graph
                              # Traffic over time
```

Analyze Menu

```
Analyze > Expert Information
                                # Automatic anomaly detection
Analyze > Follow > TCP Stream
                                # Full conversation
```



Pro Tips for Your Project

Tip 1: Color Failed Auths Red

Setup coloring rule:

1. View > Coloring Rules
2. New rule: "Kerberos Failed Auth"
3. Filter: `kerberos.error_code == 24`
4. Foreground: White, Background: Red
5. Apply

Result: Failed attempts jump out visually

Tip 2: Create Custom Columns

Add "Kerberos User" column:

1. Right-click packet list header
2. Column Preferences
3. Add new column:
 - Title: "User"
 - Type: Custom
 - Field: `kerberos.cname_string`
4. Apply

Result: See usernames at a glance

Tip 3: Save Display Filters

Save your common filters:

1. Apply filter: `kerberos.error_code == 24`
2. Click bookmark icon (next to filter bar)
3. Name it: "Kerberos Failed Auth"
4. Save

Result: One-click access to common filters

Tip 4: Export Attack Evidence

Create attack report:

1. Filter to show only attack packets
2. File > Export Packet Dissections > As Plain Text
3. Save as: `bruteforce_evidence_20250207.txt`

Use in: Portfolio documentation, reports



Analysis Workflow for Each Attack

Standard Analysis Process:

1. Open capture

```
wireshark attack_name.pcap
```

2. Quick statistics

```
Statistics > Protocol Hierarchy
```

Note: Total packets, Kerberos percentage

3. Find conversations

```
Statistics > Conversations
```

Sort by packets, identify attacker IP

4. Apply attack-specific filter

```
# Brute force
kerberos.error_code == 24

# AS-REP roasting
kerberos.msg_type == 11

# Kerberoasting
kerberos.msg_type == 12
```

5. Analyze timing

View > Time Display Format > Seconds Since Previous Packet

Calculate attack rate

6. Extract evidence

- Screenshot key findings
- Export filtered packets
- Note findings

7. Document

- Attack type
- Source IP
- Target accounts
- Timing/rate
- Success/failure

Attack-Specific Analysis Guides

Brute Force Attack Analysis

Filter sequence:

```
1. kerberos.error_code == 24
   → Count failures
```

2. `kerberos.msg_type == 11 && !kerberos.error_code && ip.src == ATTACKER_IP`
→ Check if succeeded
3. Statistics > I/O Graph
→ Visualize attack rate over time

Key metrics:

- Total attempts: (packet count)
 - Duration: (first to last packet time)
 - Rate: (attempts / duration)
 - Target user: (kerberos.cname)
 - Success: (yes/no)
-

AS-REP Roasting Analysis

Filter sequence:

1. `kerberos.msg_type == 10`
→ Count AS-REQ
2. `kerberos.msg_type == 11`
→ Count AS-REP (responses)
3. `kerberos.msg_type == 11 && !kerberos.pa_data`
→ Find vulnerable responses (no pre-auth)

Key metrics:

- Users tested: (unique cnames)
 - Roastable found: (AS-REP without `pa_data`)
 - Hashes captured: (AS-REP with `enc-part`)
-

Kerberoasting Analysis

Filter sequence:

1. `kerberos.msg_type == 12`
→ Count TGS-REQ
2. Statistics > Conversations

→ Find source IP

3. `kerberos.msg_type == 12 && ip.src == ATTACKER_IP`

→ Filter to attacker only

4. Expand each packet, note `kerberos.sname`

→ List services targeted

Key metrics:

- Service tickets requested: (TGS-REQ count)
- Unique services: (count distinct snames)
- Time window: (first to last)
- Rate: (requests / duration)



Common Issues & Solutions

Issue: "Too many packets to analyze"

Solution:

1. Apply kerberos filter first
2. Then add specific filters
3. Or: Export filtered packets to new smaller file

Issue: "Can't see Kerberos details"

Solution:

1. Check packet is UDP port 88 or TCP port 88
2. Expand packet details (click arrow)
3. If encrypted, you won't see inside (normal)

Issue: "Time format confusing"

Solution:

View > Time Display Format > Time of Day

Issue: "Lost in the data"

Solution:

1. Start with Statistics > Protocol Hierarchy
 2. Then Statistics > Conversations
 3. Identify key IPs
 4. Apply simple filters first
 5. Add complexity gradually
-



Learn by Doing: 5 Exercises

Exercise 1: Count Failed Attempts

Goal: Find how many passwords were tried

1. Open bruteforce capture
2. Filter: `kerberos.error_code == 24`
3. Note packet count at bottom

Answer format: "234 failed attempts"

Exercise 2: Find Attacker IP

Goal: Identify source of attack

1. Open any attack capture
2. Statistics > Conversations
3. TCP tab
4. Sort by Packets (descending)
5. Top IP with most packets = attacker

Exercise 3: Calculate Attack Rate

Goal: Measure brute force speed

1. Open bruteforce capture
2. Filter: `kerberos.error_code == 24`
3. Note first packet time
4. Note last packet time
5. Count packets
6. Calculate: `packets / (last - first)` in seconds

Answer format: "1.7 attempts per second"

Exercise 4: Find Compromised Accounts

Goal: See if attack succeeded

1. Open capture
2. Filter: `kerberos.msg_type == 11 && !kerberos.error_code`
3. Note cname (username)
4. If from attacker IP = compromised

Exercise 5: Extract Service List

Goal: List all services in Kerberoasting

1. Open kerberoasting capture
2. Filter: `kerberos.msg_type == 12`
3. For each packet, expand:
`Kerberos > tgs-req > sname`
4. Note each unique service

Answer format:

```
Services targeted:  
- HTTP/webserver.lab.local
```

- CIFS/fileserver.lab.local
- MSSQL/dbserver.lab.local

Quick Reference Card

Print and keep handy:

Wireshark Kerberos Quick Reference

ESSENTIAL FILTERS:

kerberos	# All Kerberos
kerberos.error_code == 24	# Failed auth
kerberos.msg_type == 10	# AS-REQ
kerberos.msg_type == 11	# AS-REP
kerberos.msg_type == 12	# TGS-REQ

ESSENTIAL MENUS:

Statistics > Conversations	# Find attacker IP
Statistics > Protocol Hierarchy	# Traffic summary
Analyze > Follow > TCP Stream	# Full conversation
File > Export Packet Dissections	# Save evidence

ATTACK PATTERNS:

Brute Force: Many error_code 24 from one IP
AS-REP Roast: Multiple msg_type 11 without pa_data
Kerberoast: Many msg_type 12 from one IP

You've Mastered Wireshark When...

Competency checklist:

- ☐ Can open a pcap file
- ☐ Can apply basic Kerberos filter
- ☐ Can find failed authentication attempts

- [] Can identify attacker IP address
- [] Can count packets in a capture
- [] Can follow a TCP stream
- [] Can export evidence to text
- [] Can create screenshots for portfolio
- [] Can calculate attack rate
- [] Can identify attack type from patterns

That's the 20% that matters.

Integration with Your Project

Complete Workflow Example:

Day 1: Capture Attack

```
sudo tcpdump -i any port 88 -w bruteforce_$(date +%Y%m%d).pcap
python3 01_bruteforce.py [options]
# Stop tcpdump
```

Day 2: Analyze Attack

```
wireshark bruteforce_20250207.pcap

# Apply filters, take screenshots, export evidence
```

Day 3: Document Attack

```
## Brute Force Attack Analysis

**Date:** 2025-02-07
**Capture:** bruteforce_20250207.pcap

**Findings:**
- Attacker IP: 192.168.88.10
- Target: admin@LAB.LOCAL
- Failed attempts: 234
- Duration: 3m 45s
```

```
- Rate: 1.04 attempts/second
- Result: SUCCESS (password: Spring2025!)

**Evidence:** [screenshots/]
**PCAP:** [captures/bruteforce_20250207.pcap]
```

Additional Resources

Official Wireshark:

- User Guide: Help > Contents
- Display Filters: <https://wiki.wireshark.org/DisplayFilters>

Kerberos-specific:

- RFC 4120: Kerberos Protocol Specification
 - MIT Kerberos Docs: <https://web.mit.edu/kerberos/>
-

Next Steps

1. Install Wireshark (if not already)

```
bash sudo apt install wireshark
```

2. Practice with your captures

- Open each attack type
- Apply the filters from this guide
- Follow the analysis workflows

3. Create screenshots

- Document each attack type
- Save evidence for portfolio

4. Integrate with monitoring

- Cross-reference with performance metrics
 - Correlate alerts with packet captures
-

You're now competent in Wireshark for Kerberos security analysis.

Time investment: 45 minutes reading + 2 hours practice = **Professional competence**

Created by: Emerson

Part of: Enterprise Kerberos Security Project

Pairs with: tcpdump guide

Next: Document your findings!